DATA ANALYTICS CAPSTONE PROJECT

Fatal Police Shootings in the United States

2015 – 2024

Sector	Public Safety / Criminal Justice Analytics
Team ID	G8
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1 Executive Summary

1.1 Problem

Fatal police shootings in the United States represent a significant public safety and civil rights challenge. Despite growing public interest and media coverage, data-driven analysis of who is being shot, under what circumstances, and by which agencies remains underdeveloped. Policy makers, law enforcement agencies, and civil society organisations lack clear, evidence-based tools to understand trends and guide reforms.

1.2 Approach

The team performed an end-to-end data analytics project using two raw datasets — one covering individual fatal shooting incidents (2015–2024) and another profiling law enforcement agencies. The pipeline included the following stages:

1. **Extraction** — Loading raw CSV data for shootings and agencies.
2. **Cleaning** — Dropping irrelevant columns, standardising types, mapping race codes, and fixing multi-agency IDs.
3. **EDA** — Exploratory visual analysis of demographics, geography, threat levels, and temporal trends.
4. **Statistical Analysis** — Advanced pattern analysis including regression, correlation, and group comparisons.
5. **KPI Engineering** — Deriving age groups, unarmed flags, time intelligence columns, flee-status grouping, and agency volume tiers.
6. **Final Merge & Load** — Joining the two datasets and exporting a Tableau-ready final CSV.
7. **Dashboarding** — Building interactive Tableau dashboards with executive and operational views.

1.3 Key Insights

Summary of Findings

- Over 10 years, fatal shootings have remained stubbornly persistent with no significant downward trend.
- Males account for the overwhelming majority (>95%) of victims.
- Black and Hispanic individuals are disproportionately represented relative to population share.
- A substantial share of victims were unarmed or armed only with a toy/replica weapon.
- Mental illness was documented in a significant portion of incidents.
- Body camera usage remains inconsistent across agencies.
- High-volume shooting agencies are concentrated in specific states.

1.4 Key Recommendations

R1. Mandate consistent body-camera use across all agencies.

R2. Expand mental health co-responder programmes to reduce lethal force in mental health crises.

R3. Strengthen de-escalation training, especially for agencies with high shooting volumes.

R4. Invest in early-warning data systems for pattern detection at the agency level.

R5. Improve demographic data collection quality to reduce "Unknown" race entries.

2 Sector and Business Context

2.1 Sector Overview

This project sits at the intersection of **public safety**, **civil rights**, and **government accountability**. Fatal police shootings are a highly charged social issue in the United States, drawing attention from legislators, advocacy groups, journalists, and the general public. Data journalism outlets such as *The Washington Post* have tracked these incidents since 2015, creating a rare publicly available longitudinal dataset.

2.2 Stakeholders

2.3 Why This Problem Matters

Each fatal shooting represents a life lost, a family affected, and — in many cases — a community's trust in policing diminished. Beyond the human cost, shootings carry legal, financial, and reputational consequences for agencies. A data-driven approach

Table 1: *Key Stakeholders and Their Roles*

Stakeholder	Role
Law enforcement leadership (chiefs, commissioners)	Operational reforms, training programmes
City and county government officials	Policy and budget decisions
Civil rights organisations	Advocacy, litigation, accountability
Federal oversight bodies (DOJ, Congress)	Legislative and regulatory action
Academic researchers	Policy impact evaluation

allows stakeholders to move beyond anecdote and ideology to evidence-based reform.

3 Problem Statement and Objectives

3.1 Formal Problem Definition

> *Using publicly available data on fatal police shootings in the United States from 2015 to 2024, how can we identify patterns in victim demographics, agency behaviour, temporal trends, and situational factors to inform targeted policy interventions?*

3.2 Scope

- **Geographic scope:** All 50 US states and the District of Columbia.
- **Temporal scope:** January 2015 – December 2024 (approximately 10 years).
- **Unit of analysis:** Individual fatal shooting incidents linked to law enforcement agencies.
- **Out of scope:** Non-fatal shootings, use-of-force incidents not resulting in death, incidents outside the USA.

3.3 Success Criteria

- A clean, analysis-ready merged dataset exported as `final_data.csv`.
- At least 5 well-defined KPIs available as computed columns.
- An interactive Tableau dashboard covering demographic, geographic, and temporal views.
- 8–12 data-backed insights with actionable recommendations.

4 Data Description

4.1 Source Citation and Access

Table 2: *Datasets Used*

Dataset	Source	Notes
`ShootOut_RawDataset.csv`	Washington Post Fatal Force database / project-supplied data	One row per fatal shooting incident
`Agency_RawDataset.csv`	PDAP / project-supplied agency registry	One row per law enforcement agency

4.2 Key Columns

4.2.1 Shootings Dataset

Table 3: *Key Columns — Shootings Dataset*

Column	Description
`date`	Date of the incident
`name`	Victim name
`age`	Victim age
`gender`	Victim gender (male/female)
`race`	Victim race (coded: W/B/H/A/N/O)
`armed_with`	What the victim was allegedly armed with
`flee_status`	Whether the victim was fleeing
`was_mental_illness_related`	Boolean — mental illness documented
`body_camera`	Boolean — officer body camera active
`agency_ids`	Semicolon-separated list of involved agency IDs
`state`	US state of the incident

4.2.2 Agency Dataset

4.3 Data Quality Issues

- **Race coded as single letters** — required mapping to full labels (e.g., "B" → "Black").
- `race_source` **and** `location_precision` — dropped as not required for analysis.
- `oricodes` **in agencies** — dropped as not required.
- `armed_with` **missing values** — filled with "unknown" rather than dropped.

Table 4: *Key Columns — Agency Dataset*

Column	Description
`id`	Unique agency identifier
`name`	Agency name
`state`	State where the agency operates
`type`	Agency type (e.g., police department, sheriff)
`total_shootings`	Cumulative fatal shootings on record for this agency

- `age` **stored as object** — converted to numeric; non-parseable values coerced to NaN.
- **Multiple agency IDs per incident** — stored as semicolon-delimited strings; exploded into separate rows per agency for join purposes.
- `date` **stored as string** — converted to datetime.

5 Cleaning and Transformation

All cleaning steps are implemented in `notebooks/02_cleaning.py`.

5.1 Major Cleaning Steps

5.2 Assumptions Made

- Incidents with NaN race are labelled "Unknown" and retained rather than dropped.
- Missing age values are retained as NaN (not imputed).
- Each exploded row (for multi-agency incidents) is treated as a separate agency-incident link.

5.3 Output Datasets

- `Data/Cleaned/clean_shootings.csv` — cleaned long-format shootings table.
- `Data/Cleaned/clean_agencies.csv` — cleaned agencies table.

6 KPI Framework

All KPIs are computed in `notebooks/05_Final_Load.py` after merging shootings with agencies.

6.1 KPI Definitions

6.2 KPI Computation Code

Table 5: *Data Cleaning Operations*

Step	Action	Rationale
Drop irrelevant columns	Removed `race_source`, `location_precision` (shootings) and `oricodes` (agencies)	Not needed for analysis
Date conversion	`pd.to_datetime(..., errors='coerce')`	Enable time-series analysis
Age conversion	`pd.to_numeric(..., errors='coerce')`	Enable demographic analysis
Gender normalisation	`.str.lower().str.strip()`	Remove inconsistent casing/whitespace
Missing `armed_with`	`.fillna("unknown")`	Retain rows; flag unknowns explicitly
Race code mapping	Mapped W/B/H/A/N/O to full labels; remaining values set to "Unknown"	Human-readable labels for dashboards
Boolean columns	`was_mental_illness_related` and `body_camera` cast to `bool`	Ensure correct data type
Multi-agency explosion	Parsed semicolon-delimited `agency_ids`, exploded to long format, converted to numeric	Enable clean join to agency table

```python
# Age Group
bins = [0, 17, 24, 34, 44, 54, 150]
labels = ['<18', '18-24', '25-34', '35-44', '45-54', '55+']
df['Age_Group'] = pd.cut(df['age'], bins=bins, labels=labels)

# Is Unarmed
df['Is_Unarmed'] = df['armed_with'].apply(
    lambda x: True if str(x).strip().lower() == 'unarmed' else False
)

# Time Intelligence
df['Year']        = df['date'].dt.year
df['Month']       = df['date'].dt.month_name()
df['Day_of_Week'] = df['date'].dt.day_name()

# Flee Group
df['Flee_Group'] = df['flee_status'].fillna('Unknown').apply(
    lambda x: 'Not Fleeing' if x == 'not' else
```

Table 6: *Derived KPI Columns*

KPI	Definition	Column	Rationale
Age Group	Victim age bucketed into 6 bands: <18, 18–24, 25–34, 35–44, 45–54, 55+	`Age_Group`	Demographic pyramid and age-based trend analysis
Is Unarmed	True if `armed_with` == 'unarmed' (case-insensitive)	`Is_Unarmed`	Key accountability metric for reform debates
Year	Calendar year extracted from `date`	`Year`	Temporal trend analysis
Month	Month name extracted from `date`	`Month`	Seasonal pattern detection
Day of Week	Day name extracted from `date`	`Day_of_Week`	Operational scheduling insights
Flee Group	Simplified flee status: Fleeing / Not Fleeing / Unknown	`Flee_Group`	Reduces granular categories for dashboarding
Agency Shooting Volume	Tiered by `total_shootings`: Low (1–5), Medium (6–20), High (>20), Unknown	`Agency_Shooting_Volume`	Identifies high-incident agencies for oversight

```
                ('Unknown' if x == 'Unknown' else 'Fleeing')
)

# Agency Shooting Volume
def categorize_agency(total):
    if pd.isna(total):  return 'Unknown'
    if total <= 5:      return 'Low Vol (1-5)'
    elif total <= 20:   return 'Medium Vol (6-20)'
    else:               return 'High Vol (>20)'

df['Agency_Shooting_Volume'] =
    df['total_shootings'].apply(categorize_agency)
```

Listing 1: *KPI Engineering — `05_Final_Load.py`*

7 Exploratory Analysis

Refer to `notebooks/03_eda.py` and associated charts for full visual output.

7.1 Major Trends

Table 7: *Key Observations from EDA*

Observation	Detail
Consistent incident volume	Fatal shootings have occurred at a roughly steady rate year-over-year since 2015 with no sustained downward trend.
Gender disparity	Males constitute over 95% of victims across all years.
Racial disparity	Black and Hispanic individuals are represented at rates significantly higher than their share of the general population.
Unarmed victims	A meaningful proportion of victims had no weapon or carried only a toy/replica, raising serious questions about threat assessment.
Mental health nexus	A significant share of incidents involved individuals experiencing a mental health crisis.
Body camera gaps	Body cameras were not active in many incidents, limiting accountability and post-incident review.
Geographic concentration	Certain states (notably California, Texas, Florida) account for a disproportionate share of incidents, partly reflecting population size.

7.2 Segment-Level Insights

Table 8: *Segment-Level Findings*

Segment	Finding
Age group	The 25–34 and 35–44 age bands account for the largest share of victims.
Flee status	A notable portion of victims were not fleeing at the time of the incident.
Agency type	Larger urban police departments account for higher absolute volumes; sheriff offices show distinct patterns.

7.3 Visual Summaries

Refer to the Tableau dashboard and EDA notebook for:

- Bar charts of shootings by race and gender.
- Line charts of annual and monthly incident trends.
- Maps of incidents by state.
- Pie/donut charts for `Is_Unarmed`, `flee_status`, and `body_camera`.
- Heatmaps of day-of-week vs. month patterns.

8 Statistical Analysis

Refer to `notebooks/04_statistical_analysis.py` *for full statistical output.*

8.1 Methods Used

The statistical analysis notebook applies a range of descriptive and inferential techniques:

- **Descriptive statistics** — mean, median, standard deviation, and distribution analysis for key numeric fields (age, incident counts).
- **Group comparisons** — comparing incident rates and demographic composition across race, state, and agency-type groupings.
- **Correlation analysis** — examining relationships between agency size, body camera adoption, and demographic profiles.
- **Trend analysis** — year-over-year incident volume and rate changes.
- **Cross-tabulation** — combining flee status, armed status, race, and mental illness flags to identify compound risk patterns.

8.2 Results Summary

Table 9: *Key Statistical Findings*

Finding	Statistical Support
No significant decline in annual incidents	Year-over-year trend analysis shows no statistically meaningful downward trajectory.
Black victims are disproportionately unarmed compared to other groups	Cross-tabulation of `race` × `Is_Unarmed`.
High-volume agencies show distinct operational profiles	Agency volume tiers correlate with state demographics and agency type.
Incidents involving mental illness show lower body-camera usage	Cross-tabulation of `was_mental_illness_related` × `body_camera`.

8.3 Business Interpretation

- The absence of a declining trend over 10 years suggests that current de-escalation and training efforts are insufficient without systemic intervention.
- The unarmed-victim disparity across racial groups indicates that threat-assessment protocols may be applied inconsistently.
- Low body-camera usage in mental-health incidents undermines accountability in precisely the cases most likely to be contested.

9 Dashboard Walkthrough

(Tableau Public link to be inserted in `tableau/dashboard_links.md`.)

9.1 Dashboard Objective

To provide law enforcement leadership, city officials, and advocacy organisations with a single-pane-of-glass view of fatal shooting patterns that enables rapid identification of high-risk agencies, demographic disparities, and temporal trends.

9.2 Executive View

- **KPIs at a glance:** Total incidents, % unarmed, % with body camera active, % mental-illness related.
- **Trend line:** Annual incident counts 2015–2024.
- **Geographic map:** Incidents by state with colour-coded intensity.

9.3 Operational View

- **Agency drill-down:** Filter by agency name or volume tier to see that agency's incident profile.
- **Demographic breakdown:** Race, gender, and age group distributions as bar/donut charts.
- **Situational breakdown:** Flee status, armed status, and body camera use.

9.4 Filters and Interactivity

Table 10: *Dashboard Filter Options*

Filter	Options
Year	2015 – 2024 (single or range)
State	All 50 states + DC
Race	White, Black, Hispanic, Asian, Native American, Other, Unknown
Is Unarmed	True / False
Agency Volume	Low / Medium / High / Unknown
Mental Illness	Yes / No

10 Key Insights

I1. No downward trend. Fatal police shootings have remained at roughly the same annual level across the entire 2015–2024 period, suggesting policy changes have not produced measurable reductions at the national level.

I2. Overwhelming male victimisation. Over 95% of victims are male across all demographic subgroups and all years.

I3. Racial disparity in unarmed victims. Black and Hispanic individuals appear at higher proportions among unarmed victims than among victims overall, suggesting differential threat-assessment outcomes.

I4. Mental health as a key factor. A significant share of incidents involved persons in mental health crises, indicating a systemic gap in non-lethal crisis response capacity.

I5. Body camera adoption is uneven. Many incidents — especially those involving mental health — lack active body camera footage, limiting post-incident accountability.

I6. Flee status is not the majority. Contrary to common assumptions, a large proportion of victims were not fleeing at the moment of the shooting.

I7. Age peak in working years. Victims are concentrated in the 25–44 age band, representing a significant productive-age population impact.

I8. Agency volume concentration. A small number of agencies account for a disproportionate share of incidents, suggesting targeted intervention could yield systemic benefits.

I9. State-level concentration. California, Texas, and Florida together account for a large share of total incidents — partly driven by population, but also by agency culture and policy environment.

I10. Armed status ambiguity. The "unknown" armed status category is non-trivial, reflecting data collection gaps that undermine comprehensive analysis.

I11. Day/time patterns exist. Incidents are not uniformly distributed across days of the week or months, pointing to operational factors (shift patterns, weather, seasonal policing) worth investigating.

I12. Agency type matters. Different agency types (city police, county sheriff, state police) show distinct demographic and situational profiles in their incident records.

11 Recommendations

11.1 R1: Mandate Universal Body Camera Activation

Action: Federal legislation or DOJ consent decree conditions should require body cameras to be active during all enforcement encounters.

Expected Impact: Improved post-incident accountability; potential deterrent effect on use of lethal force; stronger evidentiary record for investigations.

11.2 R2: Scale Mental Health Co-Responder Programmes

Action: Expand CAHOOTS-style co-responder models nationally, routing mental health calls to clinician-police teams rather than solo armed officers.

Expected Impact: Reduction in fatal outcomes for persons in mental health crises; reduced officer risk in volatile situations.

11.3 R3: Targeted Oversight for High-Volume Agencies

Action: Use the Agency Shooting Volume KPI to identify and prioritise agencies for independent audits, training interventions, and policy review.

Expected Impact: Concentrated improvement where incident rates are highest; more efficient use of oversight resources.

11.4 R4: Standardise and Improve Data Collection

Action: Require all agencies to report complete data — including race, armed status, flee status, and body-camera activation — to a federal database.

Expected Impact: Reduced "Unknown" data gap; more robust national analysis; accountability for non-reporting agencies.

11.5 R5: Implement Evidence-Based De-Escalation Standards

Action: Adopt evidence-based de-escalation curricula (e.g., PERIL framework) and track outcomes by agency over time using this dataset.

Expected Impact: Measurable reduction in lethal force incidents over a 3–5 year horizon; improved community trust metrics.

12 Limitations and Next Steps

12.1 Data Limitations

12.2 Method Limitations

Table 11: *Data Limitations*

Limitation	Impact
"Unknown" race entries	Reduces precision of racial disparity analysis.
Self-reported armed status	Potential for post-incident reporting bias.
Missing or incomplete agency linkages	Some incidents may be mis-attributed or unlinked.
No data on non-fatal incidents	Cannot compute shooting lethality rate or analyse all use-of-force.
Voluntary reporting by agencies	Under-reporting possible for agencies not covered by Washington Post tracking.

Table 12: *Methodological Limitations*

Limitation	Impact
No population-normalised rates	State comparisons are affected by population size; per-capita rates would be more rigorous.
No causal inference	Correlational findings cannot establish causality between agency characteristics and outcomes.
No officer-level data	Cannot link incidents to individual officers or track repeat-involved officers.

12.3 Suggested Future Work

- Apply time-series modelling (ARIMA / Prophet) to forecast future incident volumes.
- Merge with agency budget data to test whether higher police spending correlates with fewer shootings.
- Geocode incidents and overlay with neighbourhood-level socioeconomic indicators.
- Track policy interventions (e.g., consent decrees, body-camera mandates) and test for pre/post differences.

13 Contribution Matrix

End of Report — Team G8 | Fatal Police Shootings in the USA 2015–2024

Table 13: *Team Contribution Matrix*

Team Member	Primary Contribution Area	Key Deliverables
Adithya,Ranvendra	ETL / Data Engineering	`01_extraction.py`, `02_cleaning.py`, raw-to-clean pipeline
Ranvendra,Ankit	EDA	`03_eda.py`, exploratory charts and visual summaries
Harshita,Vriha	Statistical Analysis	`04_statistical_analysis.py`, regression and group comparison outputs
Aryan,Chinmay	KPI Design & Final Load	`05_Final_Load.py`, KPI column definitions, `final_data.csv`
Chinmay,Harshita	Dashboarding & Reporting	Tableau dashboard, this project report